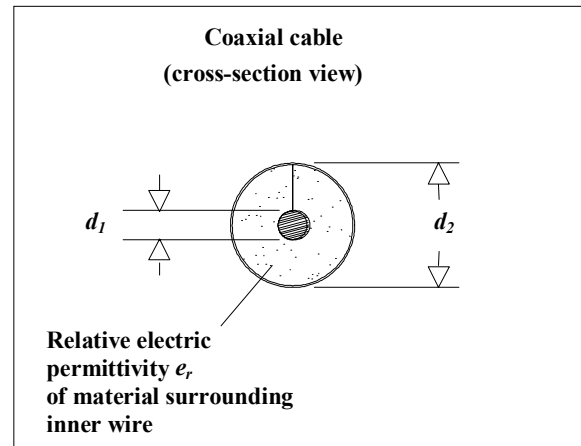


Formulas included in this spreadsheet:

Coaxial cable characteristic impedance	ZCOAX()
Coaxial cable propagation delay	PCOAX()
Coaxial cable inductance	LCOAX()
Coaxial cable capacitance	CCOAX()

Variables used:

d1	Diameter of inner wire (in)
d2	Diameter of outer shield (in)
x	Length of cable (in)
er	Relative dielectric constant of material surrounding the inner wire



coax

Characteristic impedance of coaxial cable (Ω):

$$ZCOAX(d1, d2, er) := \frac{60}{\sqrt{er}} \cdot \ln\left(\frac{d2}{d1}\right)$$

Propagation delay per in. for coaxial cable (s/in.):

$$PCOAX(er) := 84.72 \cdot 10^{-12} \cdot \sqrt{er}$$

Inductance of coaxial cable (H):

$$LCOAX(d1, d2, x) := x \cdot 5.08 \cdot 10^{-9} \cdot \ln\left(\frac{d2}{d1}\right)$$

Capacitance of coaxial cable (F):

$$CCOAX(d1, d2, er, x) := \left(\frac{x \cdot 1.41 \cdot 10^{-12}}{\ln\left(\frac{d2}{d1}\right)} \right) \cdot er$$

Example coaxial cable calculations

Diameter of AWG 30 inner wire (in.)	D1 := .01
Inside diameter of shield (in.)	D2 := .1
Length of cable (in.)	X := 20.000
Relative dielectric constant	er := 2.2

Characteristic impedance (Ω):

$$ZCOAX(D1,D2,er) = 93.144$$

Total inductance (H):

$$LCOAX(D1,D2,X) = 2.339 \times 10^{-7}$$

Same result in nH:

$$LCOAX(D1,D2,X) \cdot 10^9 = 233.943$$

Inductance per in. (H):

$$LCOAX(D1,D2,1) = 1.17 \times 10^{-8}$$

Total capacitance (F):

$$CCOAX(D1,D2,er,X) = 2.694 \times 10^{-11}$$

Same result in pF:

$$CCOAX(D1,D2,er,X) \cdot 10^{12} = 26.944$$

Capacitance per in. (F):

$$CCOAX(D1,D2,er,1) = 1.347 \times 10^{-12}$$